

Consumer Concern, Knowledge, Belief, and Attitude toward Renewable Energy: An Application of the Reasoned Action Theory

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ABSTRACT

As conventional energy resources dry up and pollution increases, utility companies and policy makers seek to develop new energy programs that are more environmentally sound and replenishable. Renewable energy has emerged as an alternative energy resource, but its marketing has not been successful despite widespread consumer concern about the environment. Using the theory of reasoned action as a theoretical framework, the current research investigates the relationship of three variables (concern with the environment, knowledge about renewable energy, and beliefs about salient consequences of using renewable energy) with consumer attitude toward paying a premium for renewable energy. As predicted by the theory of reasoned action, a positive relationship between beliefs about salient consequences and attitudes toward paying more for renewable energy was found. Overall concern levels were quite high for consumers in the sample, whereas knowledge levels were relatively low. Interestingly, it was found that consumer concern failed to translate into heightened knowledge about renewable energy. The study suggests that the consumer's

environmental concern and beliefs about renewable energy to date are more emotionally charged than fact- or knowledge-based. Implications for marketing strategies and consumer education are provided. © 2000 John Wiley & Sons, Inc.

The challenge of global sustainability could be the biggest commercial opportunity in recent history (Stuart Hart, quoted in Arnst, Reed, McWilliams, & Weimer, 1997).

Over the last three decades there has been a growing concern among consumers about whether the earth's resources will continue to sustain the standard of living that is enjoyed by most Americans (Carlson, Grove, Kangun, & Polonsky, 1996; Kilbourne, McDonagh, & Prothero, 1997; Zinkhan & Carlson, 1997). One of the major sources of consumer concern is related to energy consumption. Presently, worldwide power generation is responsible for more pollution than any other single activity (Dunn, 1997). At the same time, the demand for fossil fuel continues to grow at an alarming rate (Brown, Kane, & Roodman, 1994).

The consumption of energy represents an alarming environmental issue to consumers, but it also offers marketers an economic opportunity. As Farhar (1996) reports, Americans are beginning to recognize the link between energy use and environmental degradation. A growing segment of consumers either reward or intend to reward organizations that address environmental concerns in their business and marketing practices, and punish firms that ignore environmental imperatives (Carlson, Grove, & Kangun, 1993). Research also suggests that socially responsible consumers view environmentally friendly products more favorably (Lecky, 1993; Matulich, Haytko, & Austin, 1995; Roper Organization, 1991).

The economic opportunities presented by consumer concern for the environment have begun to influence organizational strategies. Firms are paying attention. Recent research suggests that firms increasingly recognize that environmental issues capture the attention and influence the purchasing decisions of Americans (Hart 1995; Menon & Menon, 1997; Osterhus, 1997). Accordingly, many companies are attempting to develop products and processes that are both profitable and environmentally friendly (Arnst et al., 1997). For example, renewable forms of energy are recognized as potential alternatives for alleviating the environmental problems exacerbated by the prolonged utilization of conventional sources of energy.

Renewable energy comes from easily replenishable natural resources and technologies that are environmentally sound. Table 1 presents a definition of renewable energy as well as descriptions of some of the major types of renewable energy. Green energy programs utilizing re-

Table 1. Types of Renewable Energy.

Renewable resource technologies are defined as electricity produced from other than a conventional power source provided that a power source utilizing more than 25% fossil fuel is not included (adapted from the Policy Report on Renewables Finding to the Legislature by the California Energy Commission, 1997).

- Photovoltaic cells convert solar radiation directly into electricity. PV systems include rooftop and free-standing arrays.
- Thermal electric technologies produce electricity by concentrating sunlight onto a working fluid or engine—parabolic troughs, for instance, concentrate sunlight as much as 80-fold onto a tube at the focal line of the trough.
- Wind machines convert the kinetic energy of wind into rotational energy that drives electricity generating turbines.
- Biomass is primarily wood, wood wastes, and wood by-products, but also agricultural wastes and municipal solid waste, and is combusted to produce both heat and electricity.
- Geothermal energy is heat trapped up to 3000 ft below the surface of the earth. At present, hydrothermal energy—steam, hot water, and hot brine located within 900 ft of the earth's surface—is the primary form used commercially for generating electricity and heating buildings.
- Hydropower: Moving or falling water is used to generate electricity—a mainstay of the industrial revolution in the United States and by far the most important renewable energy technology for electricity today.

Source: Kozloff (1994)

newable forms of energy such as solar, wind, or geothermal power are more environmentally beneficial than conventional forms of energy, primarily because they can be generated with little pollution and depletion of resources. Therefore, renewable energy programs can become potentially attractive alternatives for consumers who are concerned about the environment, especially in light of the fact that 75% of Americans consider themselves to be environmentalists (Osterhus, 1997). Currently, a core market for green products could represent as many as one fifth of all consumers (Ottman, 1997). Thus, it is not surprising that 23 electric power utilities across the U. S. are adding to their traditional offerings by marketing renewable forms of energy to consumers at a premium price (Ottman, 1997).

Despite widespread support for environmentally friendly products among consumers, energy corporations have not been particularly successful at marketing renewables directly to consumers (Gleason, de Alba, & Fish, 1996; Rader & Norgaard, 1996). Renewable technologies accounted for only 9% of U.S. energy consumption in 1992 and are projected not to exceed 11% by 2010 if current market trends persist (Kozloff, 1994). Three major explanations for this lackluster performance have been suggested (Ottman, 1997). The first is related to the inherent nature of this energy form, its intangibility. When consumers are unable

to observe or feel an innovative alternative to current offerings, they tend to be more reluctant to choose it unless clearly explicated characteristics or benefits override this disadvantage.

Second, because renewable energy is relatively new and is not yet widely available, consumers are usually asked to pay more for what may seem to be a similar product. Thus, perceived similarity may discourage consumers with limited discretionary income from paying a premium for renewable energy. Lastly, consumers tend to be skeptical about the misleading and overambitious environmental claims made by some green marketers. For example, little regulation or uniformity has been applied to the marketing of renewable energy, making it difficult for consumers to establish the veracity of claims made by suppliers.

The impact of these influences may lead consumers not to choose renewables unless they are properly informed of the relative advantages of renewable energy programs. Therefore, the marketability of renewable energy may be contingent on increasing consumer appreciation for the benefits of these alternative forms of energy. As the energy sector progressively deregulates, renewable energy marketers who devote more effort to creating new product offerings that address consumers' environmental concerns may take advantage of substantial emerging commercial opportunities. The need for energy suppliers to listen to the voice of the customer mandated by deregulation may lead to increased market opportunities for renewable energy.

Despite the potential impact that renewable energy programs may have on the U.S. electricity industry in the near future, empirical research that examines consumer perceptions of renewable energy is currently lacking in the marketing literature. Many of the prior studies dealing with this issue were conducted by utility companies who wanted to gauge whether consumers were willing to pay extra for renewable energy. A theoretical approach is needed to better understand sources of consumer preferences.

Prior studies have often provided conflicting information about consumers' potential behavioral reactions to renewable energy. For instance, Farhar (1996) indicates that 56–80% of respondents to national surveys say that they would pay a premium for environmental protection. In contrast, Gleason et al. (1996) suggest that less than 5% of the residential market can be expected to actually pay a premium for renewable power. Because, at least initially, renewable energy is more expensive to generate than conventional energy, a pivotal question for marketers is to determine what they can do to turn public interest in cleaner power and concern for the environment to a clear support for renewable energy programs.

The purpose of this study is to examine whether there are relationships between consumer concern for the environment, consumer knowledge and beliefs about renewable energy, and consumer willingness to pay more for renewable energy. The theoretical framework used to de-

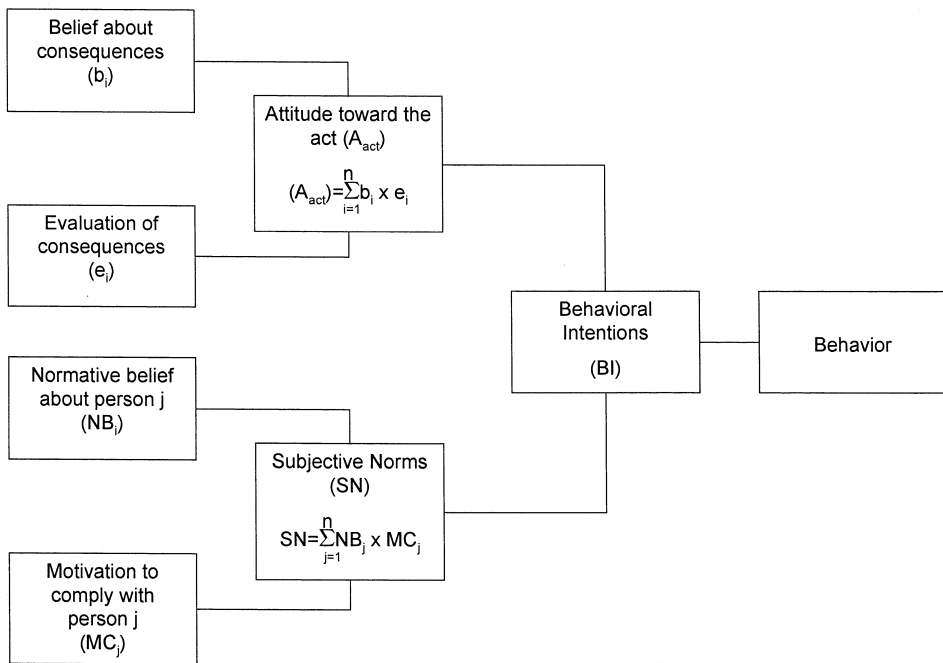


Figure 1. From Fishbein and Ajzen (1975).

velop and test our hypotheses is the theory of reasoned action (TRA) advanced by Fishbein and Ajzen (1975).

The following section presents a brief review of the theoretical framework and its application to consumer perceptions of renewable energy. Research hypotheses are formulated and the research design used to empirically test the hypotheses is outlined. Next the results of the study are described, and conclusions and implications are discussed.

THEORETICAL FRAMEWORK

The theory of reasoned action (Fishbein & Ajzen, 1975) suggests that people consider the consequences of alternative behaviors before engaging in them, and that they choose to perform behaviors they associate with desirable outcomes. In the model, behaviors are determined by a person's intention to perform the behavior. Fishbein and Ajzen suggest that behavioral intent (BI) is derived from two factors: (1) attitude toward the behavior, and (2) subjective norms (or perceived social pressure associated with the behavior). A diagram of the TRA is shown in Figure 1.

This article will focus primarily on the process leading to attitude toward the behavior, because our study surveyed consumers on their

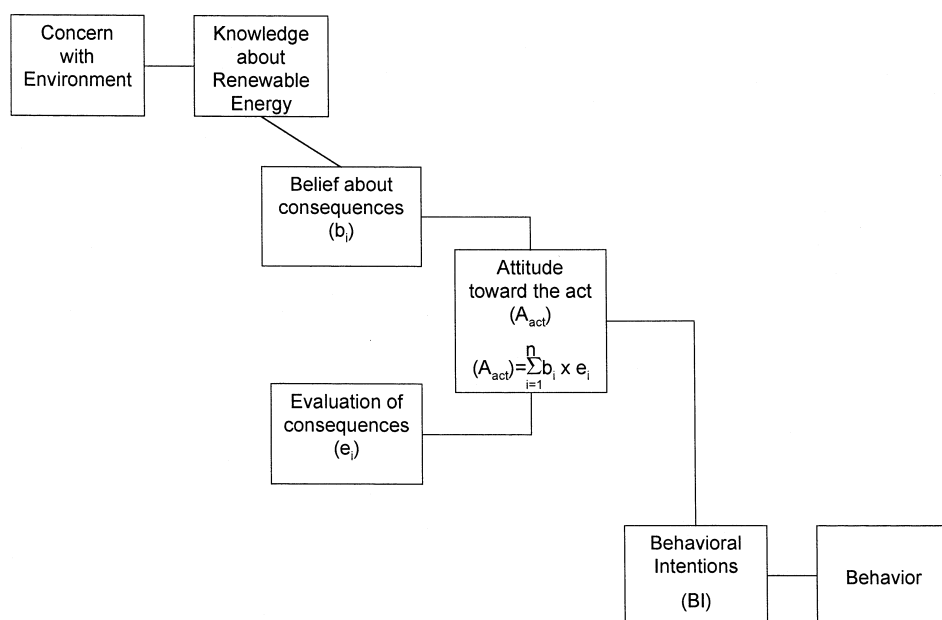


Figure 2. Theoretical framework.

attitudes toward renewable energy and their intention to engage in the behavior of paying additional monies to consume renewable energy. According to TRA, attitude is derived from two factors: a group of beliefs that one holds about the object of the behavior (b_i), and valenced evaluations (e_i) of the beliefs. Beliefs are based on knowledge, or that which the individual perceives to be true. Specifically, TRA posits that attitude toward the behavior (A_{act}) is the summation of the product of the beliefs that the behavior leads to salient consequences and the evaluation of these salient consequences ($b_i e_i$).

The other component of attitude formation in the TRA is the subjective norm. This side of the model suggests that the subjective norm associated with a behavior (B) is the sum of the product of the belief that relevant others think the individual should comply with the behavior and the motivation to comply with relevant others. Although our data are not designed to test this proposition, suggestions will be made for future research based on the subjective-norm component of the model.

The key variables measured in our study are concern for the environment, knowledge about renewable energy, beliefs about renewable energy, and willingness to pay more for renewable energy. For the concern and knowledge variables, it is our contention that these variables contribute to the belief component of the Fishbein model. Figure 2 shows the direction of the hypothesized relationships. The logic behind them is discussed in the next section.

HYPOTHESES

Consumer Concern's Relationships with Knowledge and Willingness to Pay More

It has been suggested that global warming and other environmental concerns will lead to a resurgence of interest in renewable energy (Joskow, 1996). Consumers are interested in the environment largely because they are concerned about their own health or their children's future (Esmond, 1997; Howarth & Norgaard, 1995). According to Zaichkowsky (1986), when an issue involves personal importance or values such as health or children's future, it enhances involvement, which leads to information search. Information search may, in turn, result in higher levels of consumer knowledge about alternative solutions to environmental problems such as renewable energy. Accordingly, we would expect that those people who are more concerned about the environment are more likely to be knowledgeable about renewable energy or other environmentally friendly alternatives to conventional forms of energy.

- H1:** Consumers who are more concerned about the environment tend to be more knowledgeable about renewable energy than consumers who are not as concerned about the environment.

In addition, research on the public's environmental perceptions suggests that many consumers have favorable attitudes toward environmentalism and consider themselves to be environmentalists. Consumers who are concerned about the environment indicate that they would prefer environmentally friendly products and would reward environmentally conscious marketers whom they perceive as socially responsible. With regard to the energy usage problems addressed in this study, researchers report high public preference for cleaner power. For example, Farhar (1996) evaluated data from 700 surveys conducted in the United States over the past 18 years and found that a strong majority of consumers prefer renewable energy over their current sources.

In sum, the literature suggests that consumers are prepared to financially support their concern for the environment. Therefore, it seems reasonable to propose that consumers who are more concerned about the environment are more willing to pay a premium for renewable energy than consumers who are not as concerned about the environment. Thus, we also expect the following with regard to the relationship between consumer concern for the environment and consumer willingness to pay more for renewable energy:

- H2:** Consumers who express a higher level of concern about the environment are more likely to be willing to pay more for renewable

energy than consumers who express a lower level of concern about the environment.

Knowledge and Beliefs

Perceptions and preferences about energy and the environment are influenced by objectively factual information and sometimes by a lack of it (Farhar, 1996). In the context of TRA, it makes sense that heightened knowledge about renewable energy would lead to stronger beliefs about the positive consequences or benefits of the use of renewable energy and to more positively valenced beliefs about the impact of the use of renewable energy. As indicated in Figure 2, we predict that concern affects knowledge, which, in turn affects beliefs. Thus:

- H3:** Consumers who are more knowledgeable about renewable energy tend to have stronger beliefs about the positive consequences of using renewable energy than consumers who are less knowledgeable about renewable energy.

Beliefs and Willingness to Pay More

The theory of reasoned action proposes that A_{act} is formed by the sum of $b_i e_i$. In the context of this study, the relevant measure of A_{act} is the willingness to pay more for renewable energy, because self-reported willingness to pay more reflects an attitude toward the behavior of consuming renewable energy. The questions used to measure willingness to pay more in this study did not ask about consumer intent to pay more, but rather focused on consumers' attitudes toward paying more for renewable energy.

As depicted in Figure 2, according to the TRA, $b_i e_i$ directly influences A_{act} . Hence, in the context of our study, the TRA would predict that positive beliefs about renewable energy are associated with willingness to pay more for it. Thus:

- H4:** Consumers who have stronger beliefs about the positive consequences of using renewable energy tend to be more willing to pay a premium for renewable energy than consumers who have weaker beliefs about the positive consequences of using renewable energy.

Knowledge and Willingness to Pay More

Consumers who are more knowledgeable about the problems associated with the deteriorating environment as well as the advantages of using renewable energy may have more positive attitudes toward renewable energy. Ultimately, such consumers may be inclined to engage in behavior that is consistent with their beliefs or knowledge.

In fact, a recent study indicates that consumers' willingness to pay more for environmentally friendly sources of energy increased as they became more aware of the relative advantages of renewable energy in comparison to conventional sources of energy like coal (Kozloff, 1994). In other words, consumers' positive attitude toward paying more for renewable energy may be further strengthened by their knowledge of its benefits. In support of this premise, Fazio, Powell, and Herr (1983) suggest that people's attitude toward an object is more likely to be consistent with their behavior if they encounter benefits from their own direct experience. In contrast, consumers who are less knowledgeable about renewable energy may be less inclined to justify the extra expense associated with its use. Thus, Hypothesis 5 proposes that consumers who are more knowledgeable about renewable energy tend to be more willing to pay a premium for it.

- H5:** Consumers who are more knowledgeable about renewable energy tend to be more willing to pay more for renewable energy than consumers who are less knowledgeable about renewable energy.

METHODOLOGY

Survey Instrument Development and Administration

A mail survey was employed to examine the hypothesized associations. The survey instrument was developed to gather information on respondents' concern for the environment, knowledge about renewable energy, beliefs about renewable energy, and willingness to pay more for renewable energy. A preliminary instrument was developed with the help of a focus group of electricity consumers. The initial list of items developed from the focus group was submitted to an electric utility's research division for feedback on the wording, presentation, and face validity of items in the instrument. The resulting items were then included in a preliminary instrument that was administered to a sample of 125 electricity consumers (payers). The pretest data were subjected to factor and reliability analyses. The reduced list of items was used in the final instrument.

A sample of 2600 residential electric bill payers was provided by a local electric utility in a large city in the southwest. This random sample was stratified on ZIP code and electricity usage to be representative of the customer base. The survey was administered in May of 1997, and resulted in a response rate of 13.4%, with 347 usable questionnaires included for analysis. This response rate compares favorably with previous surveys conducted by the electric utility company in this region. Descriptive statistics for the sample are provided in Table 2.

Analysis of nonresponse bias was performed by comparing early versus late responses, as recommended by Armstrong and Overton (1977).

Table 2. Descriptive Statistics for the Sample.

Characteristics	Sample	
	N	% ^a
Gender		
Male	202	58
Female	141	41
	343	99
Age		
20–29 yrs.	38	11
30–39 yrs.	80	23
40–49 yrs.	81	23
50–59 yrs.	51	15
Over 60	77	22
	327	94
Annual household income		
Below \$20,000	94	27
\$20,000–\$29,999	60	17
\$30,000–\$39,999	52	15
\$40,000–\$49,999	34	10
\$50,000–\$59,999	33	10
\$60,000 or more	47	14
	320	93
Educational attainment		
Did not finish high school	40	12
High school graduate	49	14
Some college	99	29
Bachelor's degree	82	24
Masters or higher	57	16
	327	95

^aSample % was calculated with the use of total number of respondents to survey (N = 347). Numbers do not add up to 100% due to missing cases.

The responses provided by the last quartile of respondents (those considered to be most similar to nonrespondents) were compared to responses provided by the first three quartiles of respondents. The comparison of group mean responses to survey items revealed no significant differences (at $p < .05$) for the 18 variables analyzed. Accordingly, non-response bias was not considered to be a problem.

Data Analysis

Item-to-total statistics were examined for the four scales representing concern for the environment, knowledge about renewable energy, beliefs about renewable energy, and willingness to pay more for renewable energy. Reliability coefficients of 0.87 (concern), 0.76 (knowledge), 0.80 (beliefs), and 0.86 (willingness to pay more for renewable energy) were

Table 3. Cronbach Alphas and Scale Items for Constructs.

	Cronbach Alpha
<i>Knowledge (3 items)</i>	0.76
How familiar are you with renewable energy sources?	
How familiar are you with wind-generated electricity?	
How knowledgeable are you about renewable energy?	
<i>Concern about the environment (6 items)</i>	0.87
How concerned are you about the environment (air, water, and land use)?	
How concerned are you about pollution?	
How concerned are you about water and air pollution in your city?	
How concerned are you about water usage in your city?	
How concerned are you about the environment when making purchases?	
The electric company should use less expensive energy even if the cheap energy may increase environmental pollution (reverse-scored item).	
<i>Beliefs (3 items)</i>	0.80
How important is the environment when considering renewable energy?	
How important is safety when considering renewable energy?	
How important is reliability when considering renewable energy?	
<i>Willingness to pay more for renewable energy (6 items)</i>	0.86
How willing would you be to use more expensive forms of energy to reduce pollution?	
How willing would you be to support a local project to generate energy with wind-powered devices?	
How willing would you be to pay more for your electric bill if you knew the cost paid for environmentally safe electricity?	
How willing would you be to support a fuel adjustment clause in your electric bill to subsidize the cost of developing wind-powered energy?	
How willing would you be to pay more now in exchange for possibly lower electric rates in the future?	
How willing would you be to pay more for wind-powered energy?	

obtained. All scale measures were considered reliable as Cronbach alphas equal or greater than 0.70 are generally accepted as indicating adequate reliability (Churchill, 1979; Nunnally, 1978). Alphas and the scale items for the four constructs are shown in Table 3.

To test Hypotheses 1–5, respondents were classified into two groups, either low or high with respect to concern, knowledge, and beliefs about renewable energy. Using median scores, based on a summation of responses to the items in each scale, the analysis split respondents into low and high groups for concern, knowledge, and beliefs about renewable energy. For example, the range of scores on concern for the environment was 6–42 (six items, with a 7-point scale for each). A cutoff score of 37.98 (which is equivalent to 6.33, on a 7-point scale) was es-

Table 4. T-tests of Group Differences in Means on Concern, Knowledge, Beliefs and Willingness to Pay More for Renewable Energy.

		Mean	SD	<i>t</i>	<i>p</i>
Knowledge ^a by concern ^b	Low concern	3.716	1.378	1.159	0.124
	High concern	3.529	1.510		
Willingness to pay ^c by concern	Low concern	3.952	1.300	3.781	0.000
	High concern	4.523	1.374		
Beliefs ^d by knowledge	Low knowledge	6.219	1.041	.902	0.184
	High knowledge	6.115	0.976		
Willingness to pay by beliefs	Low beliefs	3.991	1.312	3.180	0.001
	High beliefs	4.486	1.352		
Willingness to pay by knowledge	Low knowledge	4.043	1.366	3.396	0.000
	High knowledge	4.566	1.350		

^a7-point scale: 1 = "very unfamiliar," 7 = "very familiar."

^b7-point scale: 1 = "unconcerned," 7 = "very concerned."

^c7-point scale: 1 = "very unwilling," 7 = "very willing."

^d7-point scale: 1 = "not important," 7 = "very important."

established for the two groups. On this basis, 125 respondents were classified as low on concern for the environment, and 188 respondents were placed in the high concern for the environment group. *t*-tests were then performed to examine whether significant differences existed between group means.

RESULTS

The following section presents the results of the analyses examining the hypothesized relationships. Results of the *t*-tests performed to examine the relationships are presented in Table 4.

H1: Concern about the Environment and Knowledge about Renewable Energy

Respondents exhibited a very high level of concern for the environment with an average of more than 6 on a 7-point scale. In other words, for this sample, consumers' level of concern about the environment appears to be extremely high. This is not unexpected, given consumers' reported interest in their own health and their children's future in addition to frequently reported consumer anxiety over the planet's ongoing environmental deterioration. In contrast, respondents' reported level of knowledge about renewable energy was quite moderate, with a range of mid-to-high 3s on a 7-point scale. Thus, in testing Hypothesis 1, we must bear in mind that consumers who are truly unconcerned about the environment were not well represented in the sample.

The median summed response for the concern scale was 37.98 on a 6–42 range of potential scores. Accordingly, 125 respondents were clas-

sified as low-concern consumers and 188 as high-concern consumers, although consumers in the low-concern group actually have moderate to high levels of concern. When a comparison was made between those more concerned about the environment (mean = 3.529) and those less concerned (mean = 3.716), no significant difference was found between the two groups ($t = 1.159, p = .124$) for knowledge about renewable energy. Thus, Hypothesis 1 was not supported. Consumers who are more concerned about the environment were not significantly more knowledgeable about renewable energy than consumers who are less concerned about the environment. As a practical matter, it does not appear that concern levels are associated with knowledge.

H2: Concern for the Environment and Willingness to Pay More for Renewable Energy

the median summed response for knowledge was 11.00 on a 3–21 range of potential scores. The low-knowledge group included 166 respondents, and 142 respondents were classified as consumers with high knowledge about renewable energy.

When a comparison was made in terms of beliefs about the positive consequences of using renewable energy between those who are more knowledgeable about renewable energies (mean = 6.115) and those less knowledgeable about it (mean = 6.219), the results did not yield a significant difference between the two groups ($t = .902$, $p = .184$). Thus, Hypothesis 3 was not supported.

This result suggests that level of knowledge about renewable energy is not significantly related to consumers' beliefs about the positive effects of using renewable energy. However, it is also possible that the relatively low overall levels of knowledge about renewable energy indicated by respondents may have contributed to this finding. There may have been an insufficient number of consumers in the sample with a high enough level of knowledge about renewable energy to influence beliefs about the efficacy of renewable energy.

Next, the two hypotheses relating to the effect of consumer attitudes on willingness to pay more for renewable energy were tested.

H4: Beliefs about Positive Consequences and Willingness to Pay More for Renewable Energy

A central tenet of the TRA is that attitude toward a behavior (in the case of this study, consumer perceptions of whether they would be willing to pay more to use renewable energy) is influenced by beliefs that the behavior leads to salient consequences (in the case of this study, consumer beliefs about the importance of the environment, safety, and reliability when considering renewable energy). Accordingly, it was expected that consumer beliefs about the positive consequences of using renewable energy would be positively associated with consumer willingness to pay more for renewable energy.

The median summed response for beliefs was 18.99 on a 3–21 range of potential scores. One hundred twenty-five respondents were classified as having less positively valenced beliefs about the consequences of using renewable energy, and 175 respondents were included in the high-beliefs group. The results revealed a significant difference ($t = 3.180$, $p = .001$) between the two groups in the hypothesized direction. That is, consumers with more strongly valenced beliefs about the positive consequences of using renewable energy (mean = 4.486) were significantly more likely to indicate that they would be willing to pay a premium to use renewable energy, than those with weaker beliefs (mean = 3.991). Therefore, Hypothesis 4 was supported.

H5: Knowledge about Renewable Energy and Willingness to Pay More for Renewable Energy

Intuitively, enhanced knowledge about the efficacy of renewable energy and an appreciation of the ramifications of society's ongoing utilization of conventional sources of energy may influence consumers to be more willing to pay a premium for renewable energy. Accordingly, the final hypothesis in our study proposed a positive association between knowledge and willingness to pay more for renewable energy. Once again, two groups were created based on respondents' median summed scores on the knowledge scale. The result of the *t*-test supports Hypothesis 5 ($t = 3.396, p = .000$). More knowledgeable consumers were found to be significantly more likely to be willing to pay a premium for renewable energy than consumers with relatively less knowledge about renewable energy.

In summary, three out of five hypotheses were supported by the current study. The findings suggest significant positive relationships between beliefs, concern, knowledge, and consumer willingness to pay more for renewable energy. However, concern for the environment was not found to be significantly associated with knowledge. Nor was knowledge about renewable energy found to be significantly associated with consumers' beliefs about the positive consequences of using renewable energy.

Our expanded version of the TRA provides a useful theoretical framework, but with some limitations. These limitations may have been influenced by the high overall levels of concern for the environment and low overall levels of knowledge about renewable energy that were indicated by the respondents to the questionnaire. It is particularly notable that the Fishbein and Ajzen original hypothesis about the relationship between $b_i e_i$ and A_{act} is supported by this study.

DISCUSSION AND IMPLICATIONS

The current research focused on consumer perceptions of renewable energy issues. The study found support for a central tenet of the theory of reasoned action in that beliefs about renewable energy were positively related to attitude toward the act of paying more for renewable energy. Additionally, concern and knowledge were both found to be positively associated with willingness to pay more. However, overall levels of concern in the study were generally very high, reflecting current consumer views toward the environment, and making the test run in this study one comparing consumers concerned with the environment versus those very concerned with the environment.

Unlike some prior studies, which suggest that consumers' concern

translates into willingness to pay more for renewable energy, the current study raises a question as to whether such an extrapolation is justifiable. In fact, the findings of this study support the contention that the potential impact of consumers' environmental concern on their attitude toward paying more for renewable energy may be limited and unstable. Compared to the very high overall levels of concern for the environment, respondents' overall willingness to pay more scores are moderate and their reported knowledge about renewable energy is somewhat low. Thus, it does not appear that concerned consumers seek knowledge and, in turn, develop cognition-based beliefs about the value of using renewable energy. This suggests that consumers' concern may not lead to increased knowledge, but is nevertheless related to a moderately favorable attitude toward paying a premium for renewable energy.

The relative lack of knowledge about renewable energy indicated by consumers in this study is consistent with findings from prior studies (i.e., Dunn, 1997; Farhar, 1996; Gleason et al., 1996; Ottman, 1997). Although consumers are highly concerned about the environment and hold strong general beliefs about the efficacy of green products, they are not particularly knowledgeable about specific environmentally friendly alternatives like renewable energy. Ottman (1997, p. 5) reports that "many consumers lack awareness of exactly what types of energy are harnessed in producing the energy that comes into their homes, much less the environmental issues associated with its production." In fact, this phenomenon is not uncommon consumer behavior, because consumers are not usually engaged in extensive search and elaborate cognitive processing (Petty & Cacioppo, 1986).

The lack of an association between concern and beliefs warrants further discussion. It appears that consumers do not feel they need to know, or do not care much about whether they have sufficient knowledge before forming an attitude leading to a behavioral intention. Our results suggest that many consumers are saying: "I do not know much about renewable energy, but I am in favor of supporting its program because I am very concerned about current environmental conditions. I also believe that renewable energy has clear advantages even though I do not really know a lot about it." It is possible that consumers' beliefs are developed through primarily an affective, rather than a cognitive, process.

An important marketing implication is that firms may not be able to successfully persuade consumers to adopt renewable energy programs by primarily appealing to consumer concern for the environment. Consumer willingness to pay more for renewable energy was found to be positively related to beliefs. Accordingly, marketing efforts aimed at strengthening consumers' beliefs, which are the building blocks for stronger and more stable attitudes, may be more effective at stimulating the behavior of purchasing renewable energy. Building up beliefs about renewable energy could, perhaps, be accomplished through promotional

materials and advertisements that appeal to consumer rationality, rather than focusing purely on consumer emotions.

Our findings further suggest that the information processing a typical consumer goes through with regard to messages related to renewable energy may be more peripheral (i.e., who makes an environmental claim or more affect-based immediate reactions) than central (i.e., relevant facts-based, more elaborate cognitive processing). According to the elaboration-likelihood model, an attitude formed through peripheral processing tends to be neither as strong nor as enduring as an attitude formed through central processing (Petty & Cacioppo, 1986).

The strong consumer beliefs reported in the current research may not be based on facts or attained through central information processing. This would suggest that promotional programs that employ appropriate central cues, such as the salient consequences of using renewable energy or specific statistics that may make the current environmental problem more tangible, immediate, and personally relevant, may be more effective in fortifying consumer beliefs than campaigns based on emotion.

Another important implication is that because consumers do not appear to know much about renewable energy but nevertheless do not hesitate to form strong beliefs about certain aspects of renewable energy, there may be great potential for consumer marketing deception in this area. Thus, it is even more crucial to control and manage the flow of accurate environmental information to consumers so that they can function as informed consumers in the marketplace.

LIMITATIONS AND FUTURE RESEARCH

The current study has several limitations. Firstly, the study did not test all of the variables included in the theory of reasoned reaction. Most significantly, the research focused solely on the attitudinal components. Social norms may be an important variable in the case of the adoption of an innovation like renewable energy. Therefore, future research should incorporate the social-norm components of the theory so that the role of social norms in the adoption of renewable energy can be better understood.

For example, a limitation of the study concerns the possibility of social desirability bias influencing the results. Prior research suggests that respondents tend to answer questions in a self-administered survey in a socially desirable way, especially when questions deal with values (Paulus, 1991; Zerbe & Paulus, 1987). Because renewable energy issues are closely associated with societal values related to the environment, it is possible that respondents answered some questions based on what they believe to be socially desirable, especially with respect to the items measuring concern for the environment and willingness to pay more for

renewable energy. Future research should examine the potential impact of social desirability bias on consumers' responses to questions about concern for the environment and willingness to pay more for environmentally friendly products.

Finally, the methodology for the study does not allow for qualitative insights on the nature of the beliefs (i.e., primarily affect-based or cognition-based) consumers have about renewable energy. It would be useful to investigate exactly what consumers know and would like to know about renewable energy through in-depth interviews or focus groups. Also, it may be important to investigate the misconceptions consumers may have about environmental issues and programs like renewable energy initiatives. Answers to these questions may help marketers better understand consumer perceptions and as a result develop more effective marketing communication strategies.

CONCLUSION

The study found that consumers' willingness to pay a premium for renewable energy was positively related to a number of preceding variables, including beliefs about the salient consequences of using renewable energy. Although consumers reported high levels of concern, they also reported relatively low levels of knowledge about renewable energy. In addition, consumers' concern and beliefs were found to be unrelated to their knowledge. These findings suggest that consumer beliefs about renewable energy may currently be driven more by affect than by cognitive processes. Thus, the current research concludes that persuasive marketing efforts should focus on building consumer knowledge through information-based campaigns that will lead to cognition and, in turn, to stronger beliefs about the salient consequences of using renewable energy.

For example, promotional campaigns that focus on the identification of local renewable energy suppliers, the comparison of renewable energy prices to the cost of conventional energy, and a clear description of the makeup of the renewable energy on offer would furnish consumers with many of the facts they may need to make an informed choice.

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